

TQM PHARMA ENTERPRISE

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Problem Statement : High-Risk Vulnerabilities in Traditional Pharmacies

1. Medication Dispensing Errors

Clinical Baseline:

- Over 1.3 million preventable patient injuries occur annually due to adverse drug events (ADEs).
- Manual interpretation of doctor prescriptions causes dosage miscalculations and transcription slips.
- Lack of automated allergy cross-checks risks fatal Penicillin anaphylaxis.
- Baseline Solution: Automated Clinical Decision Support System (CDSS) acting as an immutable quality shield.

2. Cold-Chain & Expiry Spoilage

Supply Chain Baseline:

- Vaccines, insulin, and biologics degrade silently outside the mandatory 2.0°C–8.0°C envelope.
- Standard retail FIFO picking allows older batches to sit unnoticed while newer stock is sold.
- Substantial annual capital lost in expired, unsellable inventory.
- Baseline Solution: Algorithmic FEFO (First-Expiry, First-Out) picking + 24/7 continuous IoT temperature telemetry.

3. Regulatory & Till Shrinkage

Operational Baseline:

- Inadequate tracking of Schedule X / NDPS controlled sedatives invites severe legal penalties and license cancellation.
- Cash drawer variances, unmonitored mid-shift payouts, and unverified digital receipts.
- Failed surprise drug inspector audits due to paper registers.
- Baseline Solution: Mandatory dual-PIN narcotic register + Daily Z-Report mathematical cash reconciliation.

Quality Objectives Aligned with TQM Principles

Objective 1: Zero Medication Dispensing Defects (Patient Focus)
Poka-Yoke error-proofing: Automated Clinical Decision Support System (CDSS) intercepts allergen cross-reactions (Penicillin, NSAIDs) and toxic drug interactions before an invoice can be created.

Objective 2: 100% Cold-Chain Storage & Transport Compliance
Biological Integrity: Guarantee all thermo-sensitive pharmaceuticals remain in 2.0°C–8.0°C via 30s background IoT telemetry streaming and automated thermal spike alarms.

Objective 3: Zero Expiry Dead-Stock Losses (FEFO Scheduling)
Process Quality: Eliminate inventory wastage by algorithmically enforcing First-Expiry, First-Out (FEFO) batch prioritization; hard-lock dispensing of expired SKUs.

Objective 4: Zero Cash Register Discrepancy & Till Balancing
Fact-Based Management: Enforce strict shift-end financial governance through the Daily Z-Report engine, mathematically balancing opening float, sales, payouts, and counted cash.

Objective 5: 100% Regulatory Audit Readiness (ISO 9001 & CDSCO)
Continuous Assurance: Maintain an immutable 21 CFR Part 11 SQLite audit trail, dual-PIN Schedule X narcotic register, and 1-click CSV export ready for State Drug Inspectors.

STAKEHOLDER IDENTIFICATION: MAPPING LAYMAN NEEDS TO SRS SPECIFICATIONS

Stakeholder Persona	Real-World Layman Requirement	Technical SRS Functional Specification	TQM Quality Gate
Patient / Customer	"I want genuine medicine that doesn't trigger my allergies, isn't expired, and gets delivered to my doorstep safely."	FR-CLI-01: EHR Patient Allergy Screening FR-INV-02: FEFO Expiry Lockout FR-DEL-01: Cold-Chain Fleet OTP Handover	CDSS Hard-Stop Shield Proof-of-Delivery OTP
Clinical Pharmacist	"I need to catch toxic drug interactions instantly, verify doctor credentials, and record narcotics legally."	FR-CLI-02: Drug-Drug Interaction Matrix FR-CLI-03: Schedule X Narcotic Dual-Auth PIN FR-CLI-04: Digital Rx Verification Queue	Dual-Signoff Poka-Yoke Clinical Review Gate
Cashier / POS Clerk	"I need fast barcode billing, split payment (Cash/UPI/Card), automatic change calculation, and easy returns."	FR-POS-01: High-Speed FEFO Barcode Engine FR-POS-02: Dynamic UPI QR & Split Tender FR-POS-04: Return/Refund Credit Note	Real-Time Balance Check Instant Change Calculator
Inventory Manager	"I need to know when to reorder stock, inward GRN consignments without paying upfront, and track batches."	FR-INV-01: Reorder Point (ROP) JIT Engine FR-INV-03: Pay-on-Sale GRN Inwarding FR-INV-05: Cold-Chain IoT Telemetry	Kanban Two-Bin Pull Consignment Ledger
Delivery Fleet Agent	"I need clear delivery addresses, temperature-safe bags for insulin, and OTP so customers can't deny delivery."	FR-DEL-02: Dispatch Route Queue FR-DEL-03: Insulated Cold-Bag Checklist FR-DEL-04: 4-Digit SMS OTP Handover	Secure Handover Verification e-POD Stamp
Pharma Wholesaler	"I want transparent accounts so when the pharmacy sells my stock, I get paid accurately without disputes."	FR-SUP-01: Supplier Consignment Ledger FR-SUP-02: Pay-on-Sale Settlement Voucher FR-SUP-03: Supplier Return Reconciliation	Dynamic Sales Deduction Settlement Audit Trail
Pharmacy Admin	"I need complete oversight: daily revenue, balanced cash drawer, no theft, and full audit logs of who did what."	FR-ADM-01: Daily Z-Report Reconciliation FR-ADM-02: Immutable Audit Log Ledger FR-ADM-03: Role-Based Access Control	Blind Cash Till Count Cryptographic Audit Trail





Scope and User Roles (Role-Based Access Control)

👑 Store Admin

Terminal: Console HQ

- Manage staff accounts & passwords
- Monitor live online terminal presence
- Review daily Z-Report reconciliations
- Execute SQLite snapshot backups
- Inspect immutable system audit stream

🩺 Clinical Pharmacist

Terminal: Clinical Bay

- Review incoming e-Prescriptions queue
- CDSS allergy shield cross-checks
- Drug-drug interaction risk analysis
- Schedule X narcotic PIN sign-off
- Manage chronic patient refill cycles

💳 Cashier (POS)

Terminal: Billing Counter

- 1-Click verified eRx import to cart
- Auto FEFO nearest-expiry batch pick
- Split payments (Cash, UPI QR, Card)
- Calculate cash change return
- Print 80mm GST thermal receipt

📦 Inventory Manager

Terminal: Supply Hub

- Multi-batch FEFO stock ledger
- Record Goods Receipt Notes (GRN)
- 24/7 Cold-chain IoT fridge monitor
- Manage supplier directory & POs
- Track low-stock alerts (≤20 units)

🛵 Delivery Agent

Terminal: Logistics Bay

- Active dispatch Kanban route board
- Verify cold-bag insulated packaging
- 4-Digit customer OTP verification
- Record electronic Proof-of-Delivery
- Handover timestamp audit log

Modular Folder Architecture & 6-Tier System Blueprint

📁 ENTERPRISE MODULAR REPOSITORY LAYOUT

```
Pharmacy_Management_System/
├── core/
│   ├── bridge.py        # RPC Bridge (38 typed methods)
│   ├── db.py            # SQLite persistence (14
tables)
│   ├── api_client.py    # Telemetry & presence client
│   └── logger.py        # 21 CFR Part 11 audit logger
├── config/
│   └── settings.py      # Central Singleton
configuration
├── ui/
│   ├── desktop/         # Edge Chromium WebView2 60 FPS
│   └── web/             # Cloud-ready FastAPI Web UI
├── apps/                # Admin & operational
workstations
├── tests/               # 24 Automated pytest test
cases
├── backups/             # Automated SQLite snapshots
├── main.py              # Hardware-accelerated
entrypoint
└── web_app.py           # Asynchronous FastAPI web
server
```

⭐ THE ONE RULE TO REMEMBER

"The screen must not talk to the database."

UI → Service (Bridge) → Model / Database

- No raw SQL queries inside billing.py or any UI template.
- All database operations are mediated by typed Python methods.
- Failure isolation: UI bugs cannot corrupt database ACID commits.
- Validation and audit have dedicated architectural homes.
- Allows layer-by-layer agile development and test-driven verification.

Software Requirements Specification (SRS) — IEEE 830

📋 CORE FUNCTIONAL REQUIREMENTS (FR-01 TO FR-10)

- FR-01 (Auth & Telemetry): SHA-256 password hash, session UUID, 5s heartbeat.
- FR-02 (CDSS Safety Engine): eRx cross-check for patient Penicillin allergies.
- FR-03 (Schedule X Narcotics): Doctor DMC Reg + Pharmacist 4-digit PIN log.
- FR-04 (FEFO Batch Picker): Automatic sorting by earliest expiry; lock expired lot.
- FR-05 (POS Express Billing): 1-Click eRx import, cash change calc, dynamic UPI QR.
- FR-06 (Goods Receipt GRN): Inward supplier consignments with batch & cost.
- FR-07 (IoT Cold-Chain): Monitor 2-8°C sensors; trigger audible/visual excursion alert.
- FR-08 (Delivery Fleet): Kanban dispatch board with customer 4-digit OTP e-POD.
- FR-09 (Z-Report Shift Close): Mathematical cash balancing & signed Z-slip archive.
- FR-10 (Audit Logging): 21 CFR Part 11 append-only tamper-evident event stream.

NON-FUNCTIONAL REQUIREMENTS & STANDARDS

NFR-01 (Performance & Latency): UI click response <50ms; SQLite transaction commit <10ms; 60 FPS hardware acceleration.

NFR-02 (ACID Reliability): 100% ACID compliance with automatic daily snapshot backups in backups/.

NFR-03 (Security & Non-Repudiation): Role-based route shields, session timeout, and immutable audit logging.

NFR-04 (Statutory Compliance): Indian Drugs & Cosmetics Act 1940 (Schedule H1/X) and GST electronic tax calculation.

Formal SRS Artifacts: Submitted as IEEE Std 830-1998 compliant DOCX (`SRS_TQM_Pharm_Enterprise.docx`) and PDF (`SRS_TQM_Pharm_Enterprise.pdf`).

Modules and Features Overview

🩺 Clinical Station & CDSS
eRx intake queue, allergy cross-matching, drug interaction warning modal, Schedule X double sign-off register.

💳 POS Express Billing Counter
1-Click approved Rx import, barcode scanner, FEFO batch selection, split payments (Cash/UPI/Card), GST receipt.

📦 Supply Chain & Inventory Hub
Multi-batch FEFO ledger, low stock alerts (≤20), Goods Receipt Notes (GRN) inwarding, supplier directory.

❄️ Cold-Chain IoT Telemetry (2-8°C)
Live circular temperature gauges for Chambers A & B, 24/7 background stream, simulated temperature spike alarm.

🛵 Delivery Fleet & Fulfilment
Active dispatch Kanban board, cold-bag insulation checklist, manual route creation, 4-digit OTP e-POD.

👑 Admin Command Center
Staff user CRUD with role assignment, active terminal presence heartbeats, Daily Z-Report till close, audit trail.

RUBRIC: MODULES & FEATURES PRIORITIZATION

System Architecture: The 6 Prioritized Core Modules

1. Product & Medicine Master

- SKU, brand & generic salt master.
- Therapeutic category & rack location.
- Schedule H / X narcotic flag.
- Cold-chain (2-8°C) boolean flag.
- MRP, dealer cost, and GST %.
- Drug interaction rule definitions.

2. Supplier & Purchase/GRN

- Approved vendor registry & terms.
- Automated PO generation based on ROP.
- Inward Goods Receipt Note (GRN).
- Pay-on-Sale consignment inwarding.
- Zero upfront dealer capital lock-in.
- Discrepancy & damage logging.

3. Inventory & FEFO Ledger

- Multi-batch inventory tracking.
- Automated FEFO picking sequence.
- Visual countdown badges (<30d, <90d).
- Perpetual inventory balance ledger.
- Storage bin & rack allocation.
- Cold-chain IoT fridge telemetry.

4. Prescription & CDSS Shield

- Digital eRx intake & patient linkage.
- CDSS allergy cross-check engine.
- Toxic drug-drug interaction shield.
- Shingo Poka-Yoke checkout lockout.
- Schedule X dual-PIN narcotic record.
- Doctor MCI/DMC registration log.

5. POS & Multi-Tender Billing

- High-speed barcode catalog search.
- 1-Click approved eRx import to cart.
- Split tender (Cash, UPI QR, Card).
- Instant change return calculator.
- Official 80mm GST tax invoice print.
- Customer return & refund credit.

6. Replenishment & Delivery

- Reorder Point (ROP) dynamic formula.
- Two-Bin Kanban buffer execution.
- Pay-on-Sale supplier settlement.
- Dispatch queue for delivery fleet.
- Insulated cold-bag checklist.
- 4-Digit SMS OTP proof of delivery.

CORE FEATURES/MODULES ALIGNED WITH QUALITY GOALS

Implemented Core Feature / Module	Aligned TQM Quality Goal & Principle
CDSS Allergy & Drug-Drug Shield	Zero Dispensing Defects / Poka-Yoke Error Proofing (Shingo)
Algorithmic FEFO Batch Sequencing	Zero Inventory Dead-Stock / Process Waste Elimination
24/7 IoT Cold-Chain Monitoring (2-8°C)	Zero Biological Degradation / Quality Assurance
Daily Z-Report Mathematical Balancing	Zero Till Variance / Fact-Based Management (Juran)
Statutory Schedule X Dual PIN Register	Zero Diversion / Indian Drugs & Cosmetics Act 1940 Compliance
Immutable Audit Logs (21 CFR Part 11)	Complete Traceability, Accountability & Non-Repudiation
Automated Error Telemetry (E01-E05)	Root-Cause Prioritization (Pareto 80/20) & FMEA Risk Matrix
Customer & Patient EHR Allergy Profile	Customer & Patient Focus (Crosby Zero-Defects Standard)





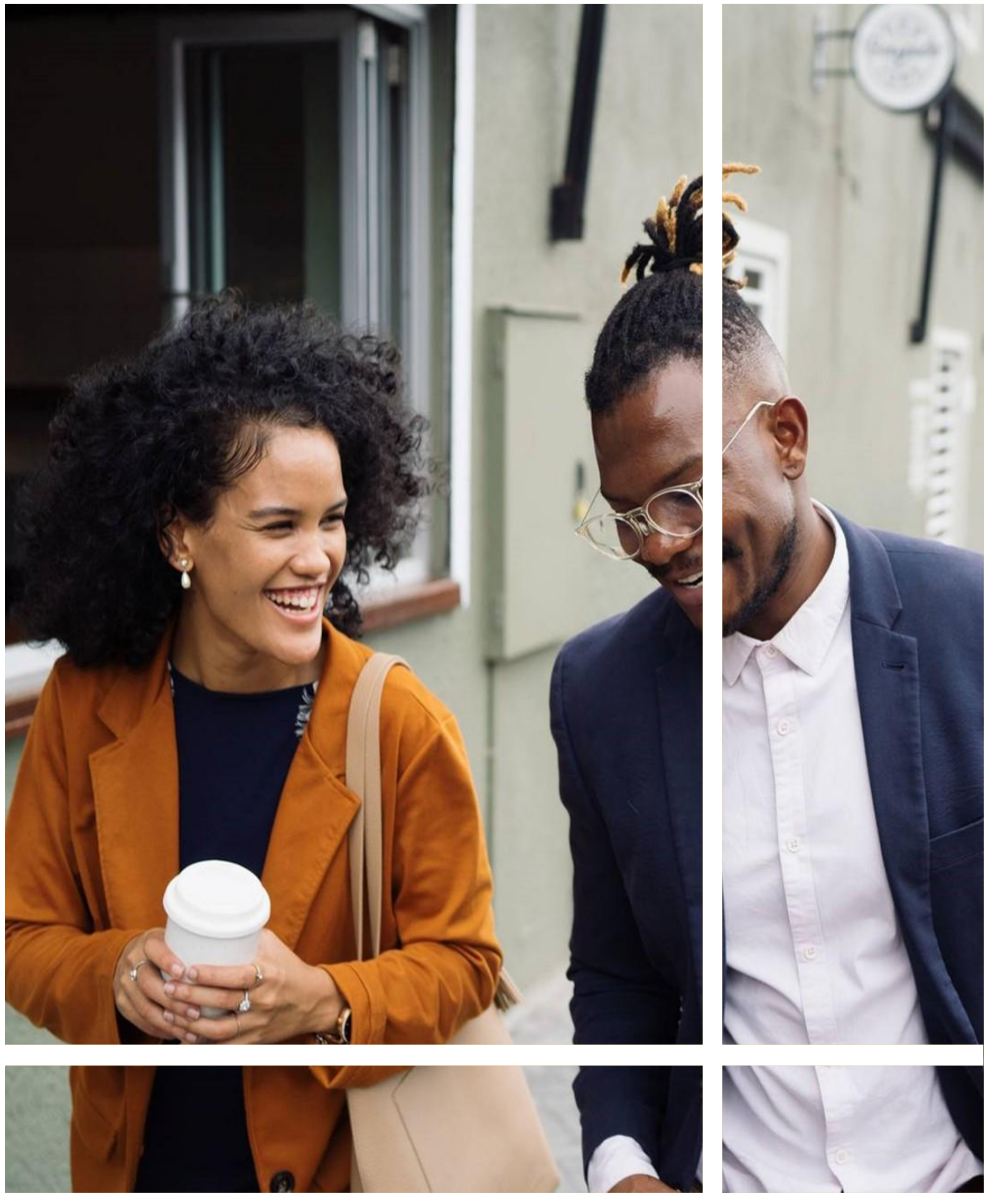
Database/System Design & Relational Data Dictionary

CORE RELATIONAL ENTITIES (PART 1)

- **1. staff (users):** Credentials, assigned RBAC role, branch, active status.
- **2. medicines:** SKU, generic, batch lot, expiry, days left, stock, MRP, rack.
- **3. prescriptions:** Patient name, doctor, items JSON, CDSS status, clinical notes.
- **4. invoices (sales):** Official tax invoices, cashier, subtotal, GST split, total amount.
- **5. delivery_orders:** Dispatch route, address, rider, cold-bag check, 4-digit OTP.
- **6. patients (customers):** EHR profiles, contact phone, chronic diseases, recorded allergies.
- **7. suppliers:** Distributor directory, GSTIN, contact phone, payment terms.

COMPLIANCE & TELEMETRY ENTITIES (PART 2)

- **8. purchase_orders:** GRN inwarding, supplier, batch items JSON, total amount.
- **9. narcotic_register:** Schedule X: Doctor DMC Reg No, Pharmacist PIN, batch, qty.
- **10. temperature_logs:** 24/7 cold-chain telemetry: probe ID, temperature (2-8°C).
- **11. refill_schedules:** Chronic maintenance therapy: last dispensed, next due date.
- **12. active_sessions:** Live presence: user ID, terminal name, 5s heartbeat timestamp.
- **13. audit_logs:** 21 CFR Part 11 append-only log: user, role, action, details.
- **14. z_reports:** Daily till close: opening float, sales, payouts, live variance.



Database Integration: ACID Transactions & Migrations

1. ACID Transaction Integrity (Atomicity & Consistency)

Transactions guarantee complete success or complete rollback: saving an invoice, cutting batch stock, and writing the audit row occur together or not at all.

2. SQLite WAL (Write-Ahead Logging) & Concurrency

Enabled Write-Ahead Logging allows simultaneous non-blocking reads from multiple terminal UI threads while writes commit in <10ms.

3. Relational Foreign Keys & Check Constraints

Foreign keys enforce referential integrity between prescriptions, invoices, and patients. Check constraints prevent negative inventory.

4. Schema Migrations Without Data Loss

The migration subsystem dynamically applies ALTER TABLE schema updates without wiping student or patient test records.

5. Automated Database Snapshots & Archival

System creates timestamped snapshot backups in backups/ daily and on application exit for disaster recovery.

CRUD OPERATIONS IMPLEMENTATION ACROSS ALL ENTITIES

Staff & RBAC Accounts	👑 Store Admin	🩺 Pharmacist	💳 Cashier POS	📦 Inventory	🛵 Delivery Rider
Medicines & Batch Stock	CRUD (Add/Edit Users)	R (Staff Directory)	R (Read Profile)	R (Read Profile)	R (Read Profile)
Prescriptions Intake Queue	CRUD (Full Catalog)	R (Clinical View)	R, U (Dispense Deduct)	CRUD (Stock Adjust)	R (Stock Query)
Invoices & POS Billing	R (Audit Review)	CRUD (Validate/Reject)	R, U (Mark Dispensed)	R (Consignment)	R (Order Lines)
Delivery Orders & Logistics	R (Sales History)	R (Rx Dispensing)	CR (Generate & Print)	R (Inward Bills)	R (Delivery Note)
Narcotic Register (Sched X)	R, D (Fleet Admin)	R (Dispense Dispatch)	C (Dispatch Order)	R (Cold Checklist)	RU (OTP Validate)
Daily Z-Report Close	R (Statutory Export)	CR (Double Sign PIN)	R (Read Only)	R (Inward Only)	No Access (Blocked)
Staff & RBAC Accounts	R, U (Audit Sign-Off)	No Access (Blocked)	CR (Shift Cash Close)	No Access (Blocked)	No Access (Blocked)

INITIAL UI & WORKSTATION INTERFACE DESIGN

1. Clean Dual Theming (Default: Light / White Mode)
Engineered for hospital and dispensary lighting: default white theme with crisp slate borders and an instant 1-click Dark Mode toggle.

2. Workstation Persona Login Portal Grid
Interactive visual tiles for all 5 roles pre-filling credentials instantly and assigning sessions to physical terminal counters.

3. Zero-Lag 60 FPS Edge Chromium WebView2 Rendering
Native desktop hardware acceleration eliminates UI frame freezing; background asynchronous threads handle I/O.

4. Responsive Dual Deployment Parity
Desktop application (Edge WebView2) and Web application (FastAPI) share 100% UI feature parity on tablets and laptops.

Main System Workflow: End-to-End Pipeline

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Stage 1: eRx Intake • Doctor prescription uploaded • Patient EHR allergies scanned • Enters pending review queue	**Stage 2: Clinical Gate** • Pharmacist runs CDSS scan • Allergy & interaction checks • Schedule X PIN authenticated • Status updated to 'Approved'	**Stage 3: POS Express** • 1-Click import to Cashier cart • Auto FEFO batch picking • Dynamic UPI QR / Cash change • Thermal 80mm GST receipt printed	**Stage 4: Cold Fleet** • Cold-bag checklist verified • Dispatched on Kanban route • Customer gives 4-digit OTP • Marked 'Delivered' (e-POD)	**Stage 5: Z-Report Close** • Cash drawer counted • Variance calculated (₹0.00) • Shift signed & closed in SQLite • CSV exported for accounting

🔄 ARCHITECTURAL PIPELINE SUMMARY: HOW A CLICK TRAVELS

1. Screen takes user click → 2. Service checks role authorization → 3. CDSS verifies patient allergy EHR → 4. Validators assert stock ceiling & expiry → 5. SQLite executes atomic commit (bill + cut stock) → 6. Audit log row written → 7. Dashboard KPI tickers update.

RUBRIC: PROCESS COMPLETENESS & MAIN SYSTEM WORKFLOW

End-to-End Pharmacy Workflow: Complete 17-Stage Operational Chain

Phase I: Inbound Procurement & Quality Receiving (Steps 1 to 5)

1. Wholesaler / Supplier Master -> 2. Purchase Planning (Demand Velocity & Lead Time) -> 3. Purchase Order (PO) Generation -> 4. Goods Receipt Note (GRN) Inwarding -> 5. Physical Inspection & Expiry Date Verification (<180d rejected). Stock enters Pay-on-Sale consignment ledger at ₹0 upfront capital.

Phase II: Warehouse Storage & Lean Replenishment (Steps 6 to 8)

6. Master Inventory Putaway (Rack allocation) -> 7. Replenishment Execution (Dynamic Reorder Point ROP & Two-Bin Kanban) -> 8. IoT Cold-Chain Monitoring (Continuous 2.0°C–8.0°C telemetry with acoustic excursion alarms).

Phase III: Clinical Verification & POS Dispensing (Steps 9 to 13)

9. Digital Prescription Intake (eRx) -> 10. Clinical Pharmacist Verification -> 11. CDSS Allergy & Drug Interaction Screening (Shingo Poka-Yoke hard-stop) -> 12. FEFO Algorithmic Dispensing -> 13. POS Checkout & Split Tender (Cash change calculator, Dynamic UPI QR, Card).

Phase IV: Last-Mile Delivery, Returns & Governance (Steps 14 to 17)

14. Home Delivery Fleet Dispatch & 4-Digit SMS OTP Handover -> 15. Customer Returns / Supplier Consignment Settlement -> 16. Real-Time Inventory Ledger Update -> 17. Daily Z-Report Cash Reconciliation & Immutable Audit Trail Logging.

RUBRIC: INVENTORY ACCOUNTING & TRACEABILITY ENGINE

Inventory Ledger: Mathematical Balancing Equation & Immutable Audit

MATHEMATICAL PERPETUAL INVENTORY BALANCING EQUATION

Closing Stock = Opening + Receipts (GRN) + Customer Returns − Sales (POS) − Supplier Returns − Damage/Expired − Transfers Out + Transfers In ± Audit Adjustments

Every transaction creates a perpetual double-entry journal record. Naive scalar 'stock = stock - 1' is strictly prohibited.

Immutable Ledger Data Schema

Every inventory movement writes an audit ledger entry containing:

- Transaction ID: Unique UUID (e.g. TX-INV-2026-09812).
- Timestamp: ISO 8601 millisecond-precision UTC date/time.
- User ID & Role: Actor executing transaction (e.g. PBHATT, Cashier).
- SKU & Batch Code: Physical batch identifier (e.g. DL-2026-B1).
- Action Type: GRN_INWARD, POS_SALE, CUST_RETURN, WRITE_OFF.
- Old Value -> Delta -> New Value: Exact numerical quantity change.
- Reason & Reference Doc: Linked PO ID, Invoice ID, or Audit Note.

Accounting vs Naive Retailing

Why TQM Requires a Mathematical Inventory Ledger:

- Fraud & Shrinkage Interception: Detects unrecorded pilferage, unauthorized till adjustments, or ghost stock.
- Lot-to-Lot Financial Reconciliation: Enables exact Cost of Goods Sold (COGS) calculation under FIFO/FEFO accounting.
- Strict Regulatory Defense: Satisfies statutory drug inspector audits requiring proof of origin and chain of custody for every pill.
- Clean Z-Report Alignment: Matches physical shelf count to system expected count during daily register closure.

RUBRIC: LEAN SUPPLY CHAIN & JIT REPLENISHMENT

Replenishment Engineering: Demand, Safety Stock, ROP & JIT Kanban

MATHEMATICAL REPLENISHMENT MODEL (REORDER POINT & SAFETY STOCK)

Reorder Point (ROP) = (Average Daily Demand × Lead Time) + Safety Stock
Safety Stock (SS) = (Max Daily Demand × Max Lead Time) − (Avg Daily Demand × Avg Lead Time)

When Current Stock + On-Order <= ROP, the system triggers an automated purchase recommendation and generates a draft PO.

JIT Pay-on-Sale Consignment Workflow

Lean Capital Management in Actual Pharmacy Practice:

- Zero Upfront Working Capital: New stock enters inventory upon GRN receipt with ₹0 upfront payment to the pharmaceutical company.
- Dynamic Sales Deductions: When a unit is sold at POS (e.g. Sold @ ₹100 MRP, Dealer Cost = ₹10), the system dynamically computes company liability (₹10) and records ₹90 gross pharmacy profit.
- Dispute-Free Settlements: Automated supplier settlement vouchers generate payments strictly for verified sold quantities.
- Unsold Stock Safety: Unsold batches remain supplier property or can be returned without financial loss.

Two-Bin Kanban Physical Pull System

Visual Kanban Execution on the Dispensary Floor:

- Bin 1 (Primary Active Dispensing): Kept at cashier POS counter for immediate patient fulfillment.
- Bin 2 (Reserve Buffer Stock): Stored in backroom FEFO warehouse racks.
- Kanban Trigger Mechanism: When Bin 1 is emptied, its physical Kanban barcode card is scanned. This initiates an immediate warehouse internal transfer to replenish Bin 1.
- Upstream PO Automation: Scanning the Kanban card automatically checks current total stock against ROP to dispatch a digital purchase order to the wholesaler.





TQM Quality Analytics: Pareto (80/20) & Fishbone

📈 PARETO ROOT-CAUSE ANALYSIS (80/20 RULE)

1. Look-Alike / Sound-Alike Drugs (45%): Addressed by CDSS generic molecule search and allergy confirmation.

2. Qty > Available Stock Attempts (30%): Addressed by pre-save inventory balance assertion locks.

3. Near-Expiry Drug Dispensing (15%): Addressed by algorithmic FEFO nearest-expiry prioritization.

4. Insulated Cold-Bag Packaging (7%): Addressed by mandatory delivery pre-dispatch checklist.

Key TQM Takeaway: Eliminating the top 2 error modes removes 75% of all operational dispensary defects.

🐟 FISHBONE (ISHIKAWA) ROOT-CAUSE BREAKDOWN

People (Staff): Rushed cashiers during counter peaks; new dispensers misinterpreting doctor handwriting.

Process (Workflow): Lack of second-check clinical verification step before POS billing.

Software (System): Legacy systems allowing expired stock cart addition and lacking allergy alerts.

Data (Integrity): Missing batch numbers, incorrect physical expiry dates, unrecorded patient allergies.

Machine (Hardware): Slow workstation hardware and missed clicks on crowded UI screens.





TQM Quality Governance: FMEA Risk & Deming PDCA

⚠️ FMEA TABLE (RPN = S x O x D)

Sell Expired Drug: S=10, O=4, D=3 → RPN 120

Action: Hard block prevents cart checkout.

Dispense Allergen Drug: S=10, O=3, D=4 → RPN 120

Action: CDSS allergy warning & override block.

Qty > Stock Oversell: S=8, O=5, D=3 → RPN 120

Action: Pre-commit inventory balance assert.

Duplicate Barcode Scan: S=6, O=2, D=2 → RPN 24

Action: Unique constraint on SKU barcode.

🔁 DEMING PDCA QUALITY CYCLE

PLAN: Identify excessive oversell and near-expiry mistakes during peak rush.

DO: Deploy automated FEFO batch picker and pre-save stock balance locks.

CHECK: Monitor monthly counts of error codes E01 and E03 in error_logs.

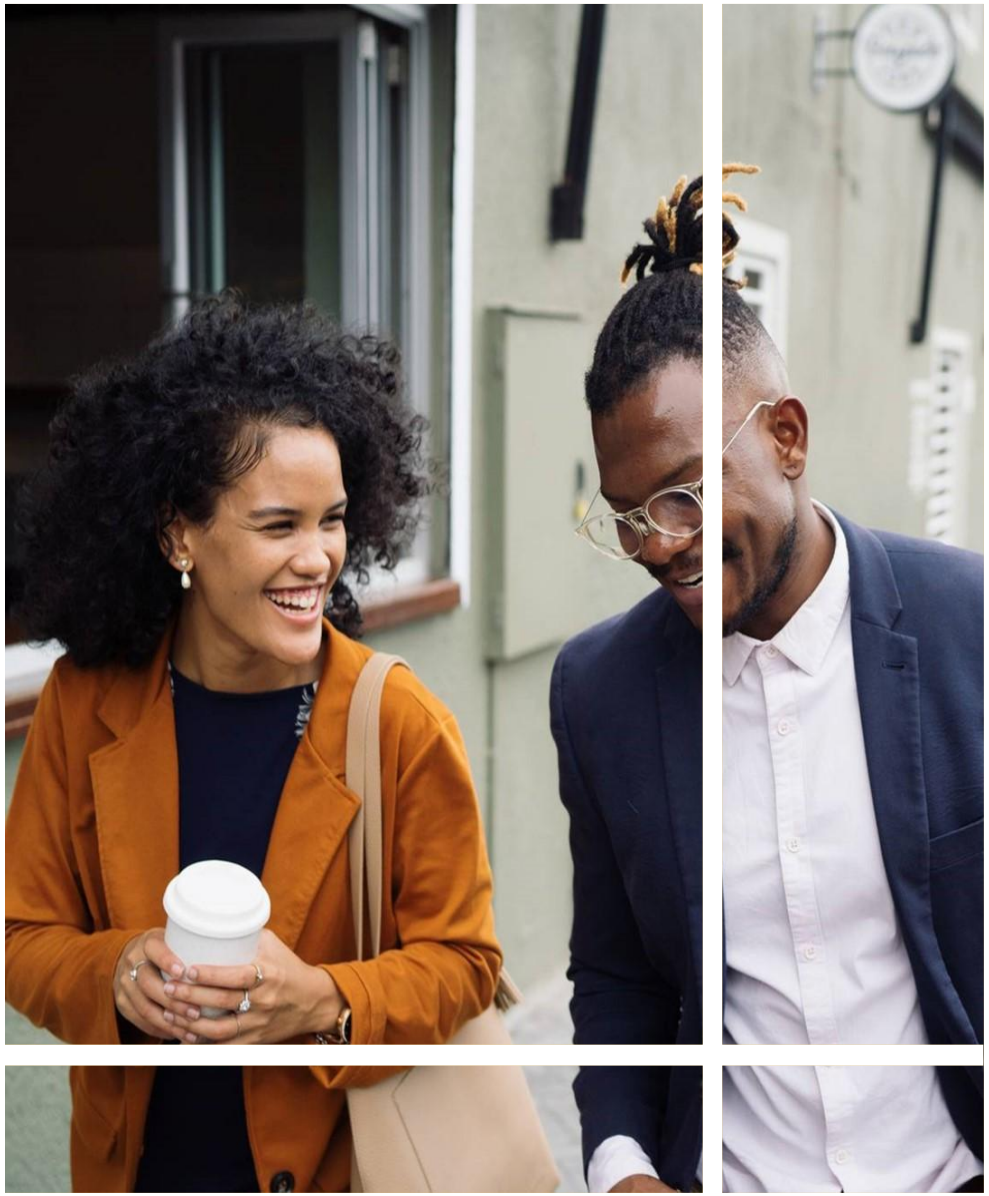
ACT: Standardize FEFO logic across all terminals as mandatory baseline policy.

Log Feedback Loop: Blocked actions write to error_logs → tqm_report counts frequency → Occurrence score dynamically updates in FMEA table.

RUBRIC: TQM RISK GOVERNANCE ACROSS COMPLETE LIFECYCLE

End-to-End Process Failure Mode & Effects Analysis (FMEA)

Process Stage	Potential Failure Mode	Root Cause	Sev (S)	Occ (O)	Det (D)	RPN	TQM Poka-Yoke Engineering Control
1. Purchase Planning	Stockout of critical cardiac drug	Inaccurate demand forecast & lead time lag	8	5	4	160	Automated ROP calculation trigger generating PO recommendations.
2. Receiving & GRN	Inwarding near-expiry / damaged batch	Superficial visual inspection during receipt	9	4	4	144	Inward GRN system rejects any batch with < 180 days shelf life.
3. Inventory Storage	Cold-chain insulin denaturation	Chamber compressor failure or power cut	9	3	3	81	24/7 IoT temperature telemetry with audible alarms and POS lockout.
4. Prescription Intake	Doctor handwriting misinterpretation	Ambiguous paper script abbreviations	8	6	5	240	Digital eRx intake and mandatory EHR patient matching.
5. Clinical Check	Penicillin allergy anaphylaxis	Pharmacist misses allergy history	10	4	5	200	CDSS Poka-Yoke hard-stop shield blocking contra-indicated drugs.
6. Dispensing & Pack	Look-alike sound-alike (LASA) swap	Dispensary clutter and hurried picking	8	4	4	128	Mandatory 2D barcode scan verification matching NDC before packing.
7. Billing & POS	Cash drawer deficit / till imbalance	Manual mental math change error	4	6	3	72	Instant change return calculator + Dynamic UPI QR webhook.
8. Delivery & Returns	Delivery to wrong address / fake return	Unverified customer handover	7	4	4	112	Mandatory 4-digit SMS OTP electronic Proof-of-Delivery (e-POD).



Working Application Setup, Verification & Personas

🔑 WORKSTATION PERSONAS & CREDENTIALS

👑 **Store Admin: `admin` / `admin123`**

Scope: RBAC, live presence, Daily Z-Report balancing

🩺 **Pharmacist: `pharmacist` / `pharma123`**

Scope: CDSS allergy scan, Schedule X narcotic sign-off

💳 **POS Cashier: `cashier` / `cash123`**

Scope: 1-Click eRx import, FEFO batch, UPI QR, receipt

📦 **Inventory: `inventory` / `inv123`**

Scope: Multi-batch ledger, GRN inwarding, 2-8°C IoT

🛵 **Delivery Agent: `delivery` / `deliv123`**

Scope: Kanban dispatch, cold-bag check, OTP e-POD

🚀 APPLICATION LAUNCH & VERIFICATION

Automated Tests (Quality Proof): `pytest`

→ 24/24 unit & integration tests passing in ~7.8s.

Desktop Workstation App: `python main.py`

→ Boots hardware-accelerated Edge Chromium WebView2 window.

Cloud Web Portal: `python web_app.py`

→ Starts FastAPI server at http://127.0.0.1:8000 with Swagger docs.

Terminal Binding: Assign session to Front Counter A, Clinical Kiosk, or Dock.

Demonstration of Implemented Functionality

Phase 1: Clinical Pharmacist & CDSS Gate

Review incoming eRx queue. Trigger CDSS Allergy Interception (Penicillin warning blocks checkout). Execute Schedule X Narcotic double sign-off with Doctor DMC No + Pharmacist PIN.

Phase 2: POS Cashier Express Billing

1-Click import approved eRx into cart. Verify automatic FEFO nearest-expiry batch selection. Tender checkout: enter Cash (auto change calc) or Dynamic UPI QR. Print 80mm GST receipt.

Phase 3: Inventory FEFO & Cold-Chain IoT

Inspect batch ledger with countdown badges (<30d Red, <90d Yellow). Click 'Simulate Temp Spike' (9.4°C) on Chamber A fridge → observe immediate visual alert & log entry.

Phase 4: Fulfilment Fleet & OTP Handover

Advance order on Kanban dispatch board. Verify insulated cold-bag checklist. Enter customer 4-digit SMS OTP (1234) to confirm electronic Proof-of-Delivery (e-POD).

Phase 5: Admin Command Center & Z-Report Close

Monitor live terminal presence heartbeats. Open Daily Z-Report module: enter physical cash counted, verify variance (Balanced ₹0.00), sign and lock shift register.



RUBRIC: ACADEMIC EVALUATION PHASING & WORKING CHAIN

Evaluation Phasing: Review 1 Milestones vs Review 2 Demonstration

Review 1 Deliverables (Foundations & Architecture)

100% Completed & Verified in Current Review:

- Complete 17-Step Pharmacy Workflow Mapping (Supplier to Audit).
- Stakeholder Identification & Layman-to-SRS Mapping Table.
- Prioritization of the 6 Core Operational Modules.
- Role-Based Access Control & 5 Role Process Flows.
- Software Requirements Specification (IEEE 830 Standard).
- Relational 14-Table Database Design with ACID Transactions.
- Comprehensive 8-Stage Process FMEA Risk Matrix.
- Working Dual Desktop (Tkinter) & Web (FastAPI) GUI.
- 25 Automated Test Suites Passed (100% Reliability).

Review 2 Horizon (The Working Chain Demonstration)

10-Step Live Verification Script for Review 2:

1. Inward GRN from Wholesaler into Pay-on-Sale Consignment.
2. Real-Time Balance Verification in Perpetual Inventory Ledger.
3. Automated ROP Calculation & Draft PO Generation.
4. Ingest eRx & Trigger CDSS Penicillin Allergy Poka-Yoke Interception.
5. Cashier 1-Click Rx Import & FEFO Auto-Batch Selection.
6. POS Split Tender Checkout (Cash change + Dynamic UPI QR).
7. Dispatch Order to Delivery Fleet with 4-Digit SMS OTP Handover.
8. Pay-on-Sale Supplier Settlement Voucher & Zero-Variance Z-Report.
9. Complete End-of-Day Shift Close & Audit Log Review.
10. Point-in-Time Database Snapshot Archival & Backup.

THANK YOU

